

Question:

There exist real numbers x and y , both greater than 1, such that $\log_x(y^x)=\log_y(x^{4y})=10$. Find xy .

Reasoning + Answer:

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<think>
1. **Understand the given equation**: We are given that there exist real numbers  $x$  and  $y$ , both greater than 1, such that :
\[\log_x(y^x) = 10 \quad \text{and} \quad \log_y(x^{4y}) = 10\]
2. **Simplify the logarithmic expressions**: Recall the logarithm power rule, which states that  $\log_b(a^c) = c \log_b(a)$ . Applying this rule to both equations :
\[\log_x(y^x) = x \log_x(y) = 10\]
\[\log_y(x^{4y}) = 4y \log_y(x) = 10\]
3. **Set up the system of equations**: From the simplified expressions, we have :
\[\begin{aligned} x \log_x(y) &= 10 && \text{(Equation 1)} \\ 4y \log_y(x) &= 10 && \text{(Equation 2)} \end{aligned}\]
4. **Interchange the variables in Equation 2**: Note that  $\log_y(x)$  can be rewritten using the change of base formula as  $\frac{1}{\log_x(y)}$ . Thus, Equation 2 becomes :
\[\begin{aligned} 4y \left( \frac{1}{\log_x(y)} \right) &= 10 \end{aligned}\]
5. **Solve for  $\log_x(y)$  in terms of  $y$ **: From Equation 1, we have :
\[\begin{aligned} x \log_x(y) &= 10 \implies \log_x(y) = \frac{10}{x} \end{aligned}\]
6. **Substitute  $\log_x(y)$  into Equation 2**: Using the result from step 5 in Equation 2 :
\[\begin{aligned} 4y \left( \frac{1}{\frac{10}{x}} \right) &= 10 \\ \text{Simplify the fraction :} \\ 4y \left( \frac{x}{10} \right) &= 10 \implies \frac{4xy}{10} = 10 \implies 4xy = 100 \implies xy = 25 \end{aligned}\]
7. **Conclusion**: The product  $xy$  is :
\[\begin{aligned} xy &= 25 \end{aligned}\]
<answer> 25 </answer>
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